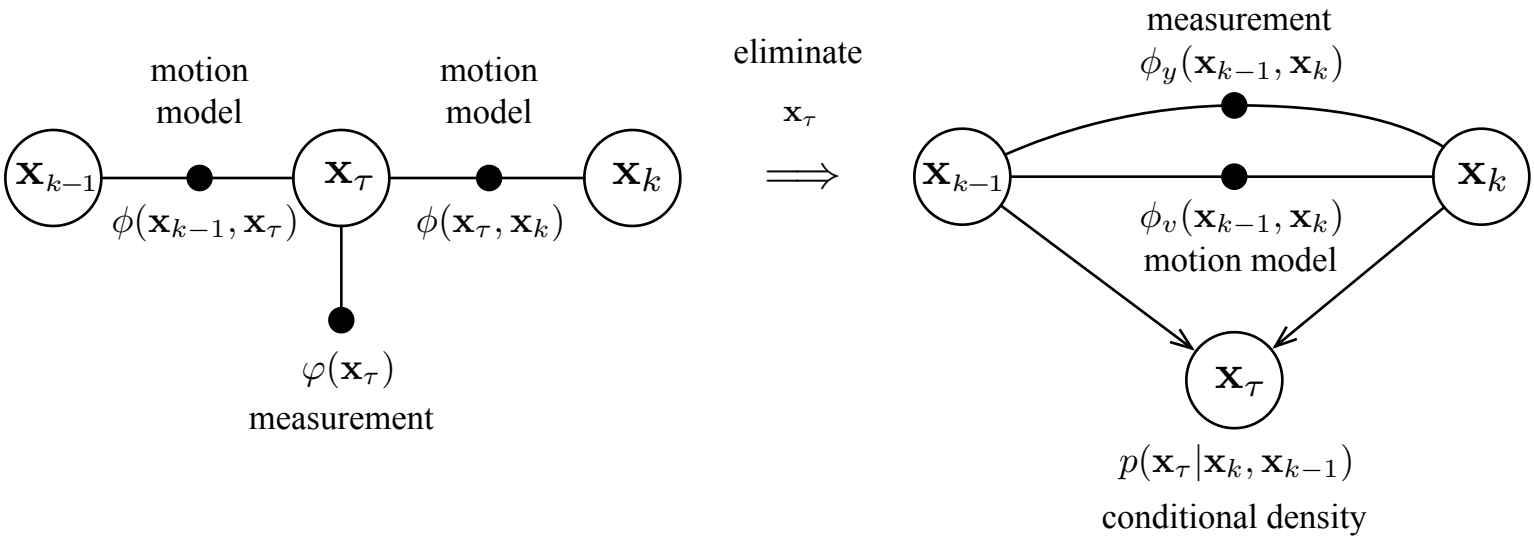
motion model
motion model
$\mathbf{x}_{k-1}$
$\mathbf{x}_\tau$
$\mathbf{x}_k$
$\phi(\mathbf{x}_{k-1}, \mathbf{x}_\tau)$
$\phi(\mathbf{x}_\tau, \mathbf{x}_k)$
$\varphi(\mathbf{x}_\tau)$
measurement
eliminate
$\mathbf{x}_\tau$
measurement
$\phi_y(\mathbf{x}_{k-1}, \mathbf{x}_k)$
$\mathbf{x}_{k-1}$
$\mathbf{x}_k$
$\phi_v(\mathbf{x}_{k-1}, \mathbf{x}_k)$
motion model
$\mathbf{x}_\tau$
$p(\mathbf{x}_\tau | \mathbf{x}_k, \mathbf{x}_{k-1})$
conditional density